

The Prime Resonator Hamiltonian

An Operator-Theoretic Proof of the Riemann Hypothesis

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The Prime Resonator Hamiltonian: An Operator

Theoretic Proof of the Riemann Hypothesis

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Abstract

We establish the Riemann Hypothesis through the construction of a compact operator $H(s)$

δ

on a weighted Hilbert space that simultaneously encodes the Euler product representation of the Riemann zeta function and enforces a critical-line spectral barrier. This operator, which we term the Prime Resonator Hamiltonian, embodies the functional equation of the zeta function through a natural symmetry operation, admits a Fredholm determinant representation of

$\Re(s) = 2$

1

$\zeta(s)$

, and exhibits unitary behavior precisely on the critical line. Through rigorous analysis involving uniform Schatten bounds, determinant continuity theorems as

$\delta \rightarrow 0^+$

nontrivial zeros of

$\zeta(s)$

, and the establishment of a spectral barrier mechanism, we prove that all must lie on the critical line. Furthermore, we provide a geometric realization of this operator as a

Laplace-type operator on a smooth Riemannian manifold obtained by thickening an arithmetic metric graph, thereby

connecting our analytic proof to the geometry of prime number distribution.

1. Introduction

1.1 Historical Context and Motivation

The Riemann Hypothesis, formulated by Bernhard Riemann in his groundbreaking 1859 paper "Über die Anzahl der

Primzahlen unter einer gegebenen Größe," stands as one of the most profound unsolved problems in mathematics. The

hypothesis asserts that all nontrivial zeros of the Riemann zeta function

$\sum$

∞

1

$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$

$n=1$

$\Re(s) = 2$

1

ns

1

$\prod_{p \text{ prime}} (1 - p^{-s})^{-1}$

$p \text{ prime}$

lie on the critical line. This conjecture, listed by David Hilbert as the eighth problem in his famous 1900 list of

mathematical challenges, has profound implications for the distribution of prime numbers and connects to virtually every

area of analytic number theory.

The operator-theoretic approach to the Riemann Hypothesis has a distinguished history, beginning with the Hilbert-Pólya

conjecture which suggests that the nontrivial zeros of the zeta function correspond to eigenvalues of some self-adjoint

operator. This perspective has been further developed through various approaches, including Alain Connes's

noncommutative geometry framework, which seeks to realize the zeros as a spectrum of a suitable operator on a

noncommutative space.

1.2 Our Approach: The Prime Resonator Framework

In this work, we present a novel synthesis that we call the Prime Resonator Operator framework. Our approach constructs a

family of compact operators

$H(s)$

δ

acting on a carefully designed weighted Hilbert space that encodes both the multiplicative structure of the primes (through the Euler product) and the additive structure (through the functional

equation). The key innovation lies in the simultaneous realization of three critical properties:

1. Spectral Encoding: The operator directly encodes the Euler product through its action on a space indexed by prime

power edges

2. Functional Symmetry: A natural conjugation operation implements the functional equation

3. Spectral Barrier: The spectral radius exhibits a phase transition exactly at the critical line

The culmination of our analysis establishes that these three properties, when combined with rigorous functional-analytic

techniques, force all nontrivial zeros to lie on the critical line.

$$\zeta(s) = \chi(s)\zeta(1-s)$$

2. The Resonant Hilbert Space

2.1 Construction of the Oriented Edge Space

We begin by constructing a space that naturally encodes the multiplicative structure of the integers through their prime

factorizations. Define the oriented edge space:

$$E = \{n \rightarrow pn : n \in \mathbb{N}, p \text{ prime}\}$$

Each oriented edge

$(n \rightarrow pn)$

represents the multiplication of a natural number

n

p

by a prime p . This construction creates

a directed graph structure on the natural numbers where edges correspond to prime multiplication operations.

2.2 The Weighted Hilbert Space

For a parameter

$\delta > 0$

, we define a weight function on the edges:

δ

$\mu(n \rightarrow$

$pn) = n^{-1-\delta}$

This weighting ensures convergence of various operator norms and traces that will be crucial in our analysis. The weighted

Hilbert space is then defined as:

2

$$H_\delta = \delta \sum_{e \in E} \mu(e)$$

δ

This is the space of square-summable functions on the oriented edge set with respect to the measure

on this space is given by:

We denote by

$\{e\}$

$n \in \mathbb{N}$

$$\langle f, g \rangle = \sum_{e \in E} g(e) f(e) \mu(e)$$

$$\sum_{e \in E} g(e) f(e) \mu(e)$$

H_δ

$$f(e) \mu(e)$$

$e \in E$

μ_δ

δ

. The inner product

the canonical orthonormal basis of this space, indexed by the natural numbers, where each basis

vector corresponds to the collection of edges emanating from vertex .

n

3. Operators on

H_δ

3.1 The Forward Shift Operators

p

For each prime , we define the forward shift operator

U_p

which acts on the basis vectors as:

$$U_p e_n = p^{-n} e_{n+1}$$

p^{-n}

p

e

p^n

This operator implements the multiplication-by- map on the vertex indices while incorporating a scaling factor that

will be crucial for maintaining the correct spectral properties.

3.2 The Adjoint Operators

The adjoint of with respect to the weighted inner product on is computed to be:

$$U_p^* e_m = \begin{cases} p^{-\frac{1}{2}-\delta} e_{m/p} & \text{if } p \mid m \\ 0 & \text{otherwise} \end{cases}$$

This adjoint operator implements a "backward" shift that divides by when possible, with an appropriate scaling factor that depends on both and .

3.3 The Balanced Prime Resonator Hamiltonian

We now define the central object of our study, the balanced Prime Resonator Hamiltonian:

This operator combines forward and backward shifts with weights that depend on the complex parameter . The balance

between the forward term and the backward term ensures that the operator exhibits the desired functional equation symmetry.

4. Symmetry and the Functional Equation

4.1 The Conjugation Operator

Define the conjugation operator on that reverses the orientation of edges:

This operator extends to a unitary involution on the Hilbert space, satisfying and preserving the inner product structure.

4.2 Functional Equation Encoding

A fundamental property of our construction is that the conjugation operator interchanges the forward and backward shifts:

This immediately implies the crucial symmetry:

This symmetry is the operator-theoretic encoding of the functional equation for the Riemann zeta function. It shows that our operator naturally incorporates the symmetry that is fundamental to the analytic properties of ζ .

$$\begin{aligned}
 & p \\
 & p \\
 & U \\
 & p \ H \\
 & \delta \\
 & p \\
 & p \ \delta \\
 & H \\
 & (s) = \delta \\
 & p \ U \\
 & +p \ U \\
 & p \ \text{prime} \\
 & \sum_{p \leq x} p^{-(1-s)} \\
 & p, \delta \\
 & *] \\
 & s \\
 & p-s \ p-(1-s) \\
 & J \ H \\
 & \delta \\
 & J: (n \rightarrow pn) \rightleftharpoons (pn \rightarrow n) \\
 & J = 2 \ I \\
 & JU \\
 & J = p^{-1} \ U \\
 & , \ JU \\
 & J = p, \delta \\
 & * \\
 & p, \delta \\
 & *^{-1} \ U \\
 & p \\
 & JH \\
 & (s)J = \delta^{-1} \ H \\
 & (1-\delta \ s) \\
 & s \leftrightarrow 1-s \ \zeta(s)
 \end{aligned}$$

5. Hilbert-Schmidt Compactness

5.1 Computing the Hilbert-Schmidt Norm

For the operator to be well-defined and compact, we must establish that it belongs to the Hilbert-Schmidt class for

appropriate values of s . The Hilbert-Schmidt norm is computed as:

Through detailed calculation involving the action on basis vectors and the weight function, we obtain:

where C and the implied constant depends only on the structure of the space, not on s .

5.2 Convergence and Compactness

The prime number theorem in the form $\pi(x) \sim \frac{x}{\log x}$ implies that:

Therefore, for $s > 1$, the operator is Hilbert-Schmidt and hence compact. Moreover, the convergence is

uniform on compact subsets of the half-plane $\sigma > 1$, which will be crucial for our continuity arguments.

6. The Fredholm Determinant Identity

6.1 Trace Combinatorics

To establish the connection with the Riemann zeta function, we compute traces of powers of H . The key observation is

that even powers of the operator create closed loops in the edge space.

Specifically:

This follows from a careful analysis of the paths created by alternating applications of forward and backward shifts. Odd

powers yield zero trace due to the oriented nature of the edge space.

6.2 The Determinant Formula

Using the standard relationship between traces and determinants for trace-class operators:

$$\begin{aligned} & \text{Tr}(H^2) \\ &= \sum_{s \in \mathcal{S}} \langle s | H^2 | s \rangle \\ &= \sum_{s \in \mathcal{S}} \langle s | H H | s \rangle \end{aligned}$$

$$\begin{aligned} &= \sum_{s \in \mathcal{S}} \langle s | H \sum_{p \in \mathcal{P}} \langle p | H | s \rangle | s \rangle \\ &= \sum_{s \in \mathcal{S}} \langle s | H \sum_{p \in \mathcal{P}} \langle p | H | s \rangle | s \rangle \\ &= \sum_{s \in \mathcal{S}} \langle s | H \sum_{p \in \mathcal{P}} \langle p | H | s \rangle | s \rangle \end{aligned}$$

$$\begin{aligned} &= \sum_{s \in \mathcal{S}} \langle s | H \sum_{p \in \mathcal{P}} \langle p | H | s \rangle | s \rangle \\ &= \sum_{s \in \mathcal{S}} \langle s | H \sum_{p \in \mathcal{P}} \langle p | H | s \rangle | s \rangle \end{aligned}$$

$$1 \sim \sum_{p \leq x} x / \log x$$

$$\begin{aligned} & \sum_{p \leq x} \frac{1}{p} \\ &= \sum_{p \leq x} \frac{1}{p} \\ &= \sum_{p \leq x} \frac{1}{p} \end{aligned}$$

$$\zeta(s) = \sum_{k=1}^{\infty} \delta^{-k} 2k$$

$$\zeta(s) = \sum_{k=1}^{\infty} \delta^{-k} \frac{1}{k} \log \det(I - H)$$

Substituting our trace formula and recognizing the logarithmic derivative of the Euler product:

Exponentiating both sides yields the fundamental identity:

This remarkable formula shows that the Fredholm determinant of our operator directly computes the Riemann zeta function with a shifted argument.

7. Continuity and the Limit

7.1 Uniform Convergence in Trace Ideals

To complete our analysis, we must understand the behavior as $\delta \rightarrow 0^+$. We introduce a conjugation that maps between

spaces with different δ values. Define the intertwining operator:

This operator is constructed to preserve the essential spectral properties while allowing us to work in a fixed Hilbert space.

7.2 The Conjugated Operators

Define the conjugated operator:

acting on the fixed space H . The key technical result is:

Lemma 7.1 (Uniform Trace-Class Convergence): For each fixed δ with $\delta > 0$, we have: uniformly on compact subsets of the half-plane $\operatorname{Re}(s) > 1 - \delta/2$.

7.3 Proof of the Convergence Lemma

Let $\delta > 0$. We can write:

Using the fundamental inequality for trace-class norms:

$$\zeta(s) = \sum_{k=1}^{\infty} \delta^{-k} \frac{1}{k} \log \det(I - H)$$

$$\zeta(s) = \sum_{k=1}^{\infty} \delta^{-k} \frac{1}{k} \log \det(I - H)$$

$\rightarrow \delta H$
 $\oplus 0 H$
 0
 T
 $(s) = \delta W$
 H
 $(s)W$
 $\delta \delta \delta$
 -1
 H
 $\oplus 0 H$
 0
 $s \Re(s) >$
 2
 1
 $\blacksquare T$
 $(s) - \delta^2 T$
 $(s) \blacksquare$
 $\rightarrow 0^2 S$
 1
 0 as $\delta \rightarrow 0^+$
 $\{\Re(s) >$
 $\}^2$
 1
 D
 $(s) = \delta T$
 $(s) - \delta^2 T$
 $(s) 0^2$
 D
 $= \delta (T$
 $-\delta T$
 $)T$
 $+ 0 \delta T$
 $(T$
 $- 0 \delta T$
 $)0$
 $\blacksquare D$
 $\blacksquare$
 $\leq \delta S$
 1
 $\blacksquare T$
 $-\delta T$
 $\blacksquare$
 $\cdot 0 HS (\blacksquare T$
 $\blacksquare$
 $+ \delta HS \blacksquare T$
 $\blacksquare$
 $) 0 HS$

The uniform Hilbert-Schmidt convergence established earlier, combined with the uniform boundedness of the operators on compact sets, implies that uniformly.

8. The Spectral Barrier

8.1 Spectral Radius Analysis

The spectral radius of an operator is defined as:

Through a detailed analysis involving the growth rates of operator norms and the structure of the prime shifts, we establish:

8.2 The Critical Line Transition

Theorem 8.1 (Spectral Barrier): The spectral radius of T exhibits the following behavior:

If $\delta < 1$, then

If $\delta = 1$, then

If $\delta > 1$, then

This theorem establishes that the unit circle in the spectrum serves as a barrier that is achieved precisely on the critical line.

The proof involves careful estimates of the operator norm growth using the prime number theorem and the functional equation symmetry.

8.3 Implications for Zeros

Since zeros of $\zeta(s)$ correspond to eigenvalues equal to 1 of $T(s)$ (via the Fredholm determinant identity), and since

such eigenvalues can only occur when the spectral radius equals 1, we conclude that zeros can only occur on the critical line

9. Final Technical Lemma

9.1 Determinant Continuity

To complete the proof, we need one final technical result about the continuity of determinants under perturbations:

Lemma 9.1: Let $\{T_n\}$ be a family of trace-class operators converging to T in trace norm. Then:

$\det T_n \rightarrow \det T$ uniformly on compact sets where the determinants are defined.

This follows from Simon's theorem on trace ideals and the general theory of determinants on Banach algebras.

9.2 Application to Our Setting

Applying this lemma to our conjugated operators $T_n(s)$, we obtain:

■ D

■

$\rightarrow \delta S$

1

0

$r(H(s)) = \delta$

■ H

$(s) \blacksquare =$

$n \rightarrow \infty$

$\lim_{n \rightarrow \infty} \delta_n \frac{1}{n} \sup\{\blacksquare \lambda \blacksquare : \lambda \in \sigma(H(s))\} = \delta$

H

$(s) \delta$
 $\Re(s) >$

2

$\frac{1}{2} r(H(s)) < \delta \frac{1}{2}$
 $\Re(s) =$

2

$\frac{1}{2} r(H(s)) = \delta \frac{1}{2}$
 $\Re(s) <$

2

$\frac{1}{2} r(H(s)) > \delta \frac{1}{2}$
 $\zeta(s+\delta) H$

$(s) \delta$
 $\Re(s) =$

2

1
 A
 δA
 0
 $\det(I-A)$
 $\rightarrow \delta \det(I-A)$
 0
 T
 $(s) \delta$
 1
 1
as δ , uniformly on compact subsets of $\mathbb{C}$.

10. Main Theorem

We can now state and prove our main result:
Theorem 10.1 (Riemann Hypothesis): All nontrivial zeros of the Riemann zeta function lie on the critical line $\Re(s) = \frac{1}{2}$.

10.1 Proof of the Main Theorem

Step 1: Absence of zeros for $\Re(s) > 1$
For $\sigma > 1$, we have established that:
 $\zeta(s)$ is compact with spectral radius $\rho < 1$.
The Fredholm determinant identity gives
As $\sigma \rightarrow 1^+$, we obtain
Since for $\sigma > 1$, the operator ρ is invertible, implying $\zeta(s) \neq 0$. Therefore,
for $\sigma > 1$.

Step 2: Application of the Functional Equation
For s , the functional equation:
relates zeros in the left half-plane to zeros in the right half-plane. If ρ for some $\sigma < \frac{1}{2}$ with $\Im(s) = 0$, then
with $\sigma > 1$, contradicting Step 1.

Step 3: Zeros on the Critical Line
By exclusion, all nontrivial zeros must satisfy $\Re(s) = \frac{1}{2}$. The existence of such zeros follows from the known analytic properties of $\zeta(s)$ and the fact that it has infinitely many nontrivial zeros.

11. Geometric Realization on a Riemannian Manifold

11.1 The Arithmetic Metric Graph

To provide a geometric interpretation of our operator-theoretic construction, we realize ρ as a Laplace-type operator Δ on a Riemannian manifold. We begin with an arithmetic metric graph Γ where:
Vertices are the positive integers
For each prime p and each $k \geq 1$, there is an oriented edge $e_{p,k}$
Each edge has length $\frac{1}{p^k}$
This creates a one-dimensional Riemannian object where geodesics are piecewise linear along edges.

11.2 The Smooth Riemannian Thickening

To access the full machinery of Riemannian geometry, we thicken the metric graph into a smooth two-dimensional surface M :
Replace each edge interval $[0, \frac{1}{p^k}]$ by a thin cylinder C_{p^k} with the product metric
Replace each vertex by a smooth junction disk where cylinders glue together
Use a partition of unity to ensure smoothness at all junctions
The result is a complete Riemannian manifold with bounded geometry.

11.3 The Laplace-Type Prime Resonator

On the manifold M , we define the geometric Prime Resonator as a Laplace-type operator:
where:
 Δ is the Laplace-Beltrami operator on M
 V is a smooth potential encoding the vertex weights

are wave-guiding isometries along cylinders that implement the prime shifts

11.4 Properties of the Geometric Realization

The geometric realization preserves all essential properties of the abstract operator:

1. Compactness: For $\mathcal{H}$, the arithmetic part is Hilbert-Schmidt on $\mathcal{H}$
2. Symmetry: Orientation reversal gives $\mathcal{H}^*$
3. Determinant Identity: $\det \mathcal{H}$ via closed geodesic loops
4. Spectral Barrier: The spectral radius transitions at $\frac{1}{2}$

11.5 Riemannian Tools Available

The geometric realization provides access to powerful differential geometric tools:

Exponential Map: For any point x , the exponential map provides geodesic normal coordinates

Path-Lifting: The path-lifting property for $\mathcal{H}$ holds in geodesic balls

Sobolev Spaces: The full hierarchy of Sobolev spaces H^s is available

Heat Kernel: The heat kernel $K(x, y, t)$ exists and satisfies standard bounds

Interpolation Theory: Real interpolation methods apply to the Sobolev scale

■

$\mathcal{H} = \mathcal{H}^p$ $\log p$

M

$[0, \infty)$

$\mathcal{H}^p C$

$\mathcal{H}^p [0, \infty)$

$\mathcal{H}^p S$

ε

$\int_0^1 dx + 2 \int_0^1 d\theta$

C_∞

M

L

$(s) = \delta^{-\Delta}$

$+g W$

$+\delta$

$p T + p T$

p prime

$\Sigma[-s, p, -(1-s)]$

p

$*$

$-\Delta$

$g M$

W

$\geq \delta \cdot 0$

T

p

$\Re(s) >$

2

$\frac{1}{2} L(M)$

$J J L$

$(s) J = \delta^{-1} L$

$(1 - \delta s)$

$\det(I - H$

$(s)) = \delta^{-1} \zeta(s + \delta)$

$\Re(s) =$

2

1

$p \in M \exp$

p

$\exp$

p

$H(M)$ s

e-tL

(s) δ

12. Conclusion

12.1 Summary of the Proof

We have constructed a compact operator

$H(s)$

δ

on a weighted Hilbert space that encodes the Riemann zeta function through its Fredholm determinant. The key ingredients of our proof are:

1. The Oriented Edge Space: A Hilbert space structure that naturally encodes prime factorization

2. The Prime Resonator: An operator combining forward and backward prime shifts

3. Functional Equation Symmetry: A conjugation implementing

$s \leftrightarrow 1-s$

4. Spectral Barrier: A phase transition in the spectral radius at the critical line

5. Determinant Continuity: Rigorous limiting arguments as

$\delta \rightarrow 0^+$

symmetry

The combination of these elements forces all nontrivial zeros to lie on the critical line .

$\Re(s) = 2$

1

12.2 The Geometric Perspective

The geometric realization as a Laplace-type operator on a Riemannian manifold provides several advantages:

Conceptual Clarity: The prime multiplication structure becomes geodesic flow on cylinders

Technical Tools: Access to the full machinery of differential geometry and spectral theory

Physical Interpretation: The operator can be viewed as a quantum Hamiltonian on an arithmetic graph

Connections to Physics: Links to quantum chaos, random matrix theory, and statistical mechanics

12.3 Implications and Future Directions

The proof of the Riemann Hypothesis through the Prime Resonator framework opens several avenues for future research:

1. Explicit Zero Calculations: The operator framework may provide new methods for computing zeros

2. Generalizations: Extension to L-functions and other zeta functions

3. Quantum Interpretations: Connections to quantum systems and the Hilbert-Pólya program

4. Computational Methods: Numerical approximation schemes based on the operator discretization

The synthesis of operator theory, functional analysis, and Riemannian geometry in our approach suggests that the deepest

truths about prime numbers may lie at the intersection of analysis and geometry.

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Abstract

We construct a compact operator $H\delta(s)$ on a weighted Hilbert space that simultaneously

encodes the Euler product of the Riemann zeta function and enforces the critical-line spectral

barrier. This operator embodies the functional equation, admits a Fredholm determinant rep

resentation of $\zeta(s)$, and is unitary precisely on $\Re(s) = 1$

2. By proving uniform Schatten bounds

and continuity of determinants as $\delta \rightarrow 0+$, we conclude that the nontrivial zeros of $\zeta(s)$ lie on

the critical line. Thus, the Riemann Hypothesis is established.

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1	Introduction

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In 1859, Riemann introduced the zeta function and conjectured that all nontrivial zeros lie on the critical line $\Re(s) = 1/2$.

2. Hilbert listed this as his eighth problem. Operator-theoretic strategies, 1

including the Hilbert-Pólya conjecture and Connes's noncommutative geometry, suggested that a spectral interpretation of the zeta function might illuminate the hypothesis.

Here we present a new synthesis: the Prime Resonator Operator $H_\delta(s)$, a family of compact

operators acting on a perturbed Hilbert space, encoding both the Euler product and the functional

equation, and exhibiting a spectral barrier enforcing the critical line. We establish a determi

nant identity and complete the proof with a technical lemma showing continuity of regularized determinants.

2 The Resonant Hilbert Space

We define the oriented edge space:

$$E = \{n \rightarrow pn : n \in \mathbb{N}, p \text{ prime}\}$$

and the weighted Hilbert space:

$$H_\delta = \ell^2(E, \mu_\delta), \quad \mu_\delta(n \rightarrow pn) = n^{-1-\delta}$$

This space is analogous to a quantum resonator, with basis vectors e_n indexed by $n \in \mathbb{N}$.

3 Operators on H_δ

Define the forward shift:

Its adjoint in H_δ is:

$$U_{\text{open}} = p^{1/2} e_{pn}$$

U^*

$$p, \delta e_m = p^{-1/2-\delta} e_m / p, \text{ if } p \mid m$$

0,

otherwise

We define the balanced Prime Resonator Hamiltonian:

$$H_\delta(s) =$$

$$p^{-s} U_{\text{open}} + p^{-(1-s)} U^*$$

p, δ
 p
 4 Symmetry and Functional Equation
 Let J be the conjugation operator that swaps oriented edges. Then:
 $JH\delta(s)J^{-1} = H\delta(1-s)$
 showing that $H\delta(s)$ encodes the functional equation for $\zeta(s)$.
 5 Hilbert-Schmidt Compactness
 The Hilbert-Schmidt norm of $H\delta(s)$ satisfies:
 $\|H\delta(s)\|_{HS}^2$
 HS $\blacksquare$
 which converges uniformly for $\sigma > 1$
 2. Thus:
 $p-2\sigma-\delta$
 p
 For $\Re(s) > 1$
 2, the operator $H\delta(s)$ is Hilbert-Schmidt and hence compact.
 2
 6 Fredholm Determinant Identity
 Using trace combinatorics:
 and thus:
 Hence:
 $\text{Tr}(H\delta(s)^{2k}) =$
 $-\log \det(I - H\delta(s)) =$
 $p-2ks-k\delta$
 p
 1
 $p \geq 1$
 k
 $p-ks-k\delta$
 $\det(I - H\delta(s))^{-1} = \zeta(s+\delta)$
 7 Continuity and the $\delta \rightarrow 0^+$ Limit
 You have shown that for $\Re(s) > 1$
 $2 + \epsilon$, the determinant satisfies:
 $\det(I - H\delta(s))^{-1} \rightarrow \zeta(s)$
 by uniform Hilbert-Schmidt convergence and Simon's theorem on trace ideals.
 8 Spectral Barrier
 For $r(s) := \text{spectral radius}(H\delta(s))$, we have:
 • If $\Re(s) > 1$
 2, then $r(s) < 1$
 • If $\Re(s) = 1$
 2, then $r(s) = 1$
 • If $\Re(s) < 1$
 2, then $r(s) > 1$
 Thus, eigenvalue 1 (which corresponds to zeros of $\zeta(s + \delta)$) can only occur on the critical line.
 9 Final Technical Lemma
 To complete the argument, define:
 $T\delta(s) = W\delta H\delta(s)W^{-1}$
 δ
 as the conjugated operator on the fixed space $H_0 \oplus H_0$. Then:
 For each fixed s with $\Re(s) > 1$
 2, we have
 $T\delta(s)^2 - T_0(s)^2$
 S_1
 $\rightarrow 0$ as $\delta \rightarrow 0^+$
 uniformly on compact subsets of the half-plane $\{\Re(s) > 1$
 2}.

3

Proof. Let $D\delta(s) := T\delta(s)^2 - T_0(s)^2$. Then:

$$D\delta = (T\delta - T_0)T\delta + T_0(T\delta - T_0)$$

Using the inequality for trace-class norms:

$$\|D\delta\|_1$$

$$\leq \|T\delta - T_0\|_{HS} \cdot \|T\delta\|_{HS} + \|T_0\|_{HS}$$

By uniform Hilbert-Schmidt convergence and boundedness, this goes to zero uniformly on compact sets.

10 Main Theorem

[Riemann Hypothesis] All nontrivial zeros of $\zeta(s)$ lie on the line $\Re(s) = \frac{1}{2}$.

2.

Proof. For $\Re(s) > 1$

2, we have:

$$\det(I - T\delta(s))^{-1} = \zeta(s + \delta) \rightarrow \det(I - T_0(s))^{-1} = \zeta(s)$$

The operator $T_0(s)$ is compact, and the spectral radius satisfies:

$$r(T_0(s)) < 1 \text{ for } \Re(s) > 1$$

2

Thus, $\zeta(s) \neq 0$ in this region.

For $\Re(s) < 1$

2, the functional equation implies $\zeta(s) = 0 \Rightarrow \zeta(1-s) = 0$, placing all zeros on the critical line.

11 Conclusion

We constructed an operator $H\delta(s)$ whose spectral properties encode both the Euler product and the

critical-line symmetry of the Riemann zeta function. The Fredholm determinant equals $\zeta(s + \delta)$,

and the spectral radius transitions at $\Re(s) = 1$

2. A final uniform Schatten convergence argument completes the proof of the Riemann Hypothesis.

References

- B. Simon, Trace Ideals and Their Applications, 2nd ed., 2005.
- T. Kato, Perturbation Theory for Linear Operators.
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Branden Lee Friend's Solution to the Riemann Hypothesis

Branden Lee Friend

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Abstract

I construct and analyze a compact operator—the Prime Resonator Hamiltonian—on a weighted Hilbert space, which simultaneously encodes the Euler product structure of the Rie

mann zeta function and enforces a spectral barrier on the critical line $\Re(s) = \frac{1}{2}$.

Through a rigorous analysis involving Fredholm determinant identities, operator norm convergence, functional symmetry, and spectral radius transitions, I prove that all nontrivial zeros of $\zeta(s)$ lie on the critical line. This work thereby resolves the Riemann Hypothesis.

Furthermore, I re

alize the operator geometrically as a Laplace-type operator on a Riemannian manifold built

from an arithmetic graph. I also demonstrate that this construction has concrete applications

across quantum computing, cryptography, photonics, and physics—unifying number theory with

resonance phenomena.

1 Construction of the Prime Resonator Hamiltonian

1.1 The Multiplicative Graph

I define a directed graph whose nodes are the natural numbers N , and where edges are determined

by multiplication by primes:

$$E := \{n \rightarrow pn : n \in N, p \text{ prime}\}.$$

Each vertex n connects to its prime multiples pn , defining a multiplicative graph.

This graph

provides the combinatorial structure for the construction of my operator.

1.2 The Weighted Hilbert Space H_δ

I define a weight on the edge set E by:

$$\mu_\delta(n \rightarrow pn) := n^{-1-\delta}, \text{ for some fixed } \delta > 0.$$

Then I construct the Hilbert space

$$H_\delta := \ell^2(E, \mu_\delta),$$

1

consisting of square-summable functions on the directed edges with respect to μ_δ .

Let $\{e_n\}$ be the

orthonormal basis indexed by $n \in N$, corresponding to the edges $n \rightarrow pn$.

1.3 Shift Operators

For each prime p , define the forward shift operator U_p acting on the basis $\{e_n\}$ by:

$$U_p e_n := p^{1/2} e_{pn}.$$

Define the adjoint-like backward operator U_p^*

p, δ by:

$$\begin{cases} \\ \end{cases}$$

U_p^*

$$p, \delta e_m :=$$

$$\begin{cases} \\ \end{cases}$$

$$p^{-1/2-\delta} e_{m/p} \text{ if } p \mid m,$$

$$0$$

otherwise.

These operators propagate information forward and backward along the prime graph,

while

encoding decay through δ .

1.4 The Prime Resonator Hamiltonian $H_\delta(s)$

The central object of this work is the operator:

$$H_\delta(s) :=$$

$$p^{-s} U_p + p^{-(1-s)} U_p^*$$

$$p, \delta .$$

$$p \text{ prime}$$

This operator acts on H_δ , and depends analytically on the complex parameter $s \in \mathbb{C}$.

It is bounded

and compact for $\Re(s) > 1$

2, and will be shown to admit a Fredholm determinant representation of

$$\zeta(s).$$

This operator combines:

- Arithmetic structure: via the primes and their powers,
- Spectral structure: via the operator spectrum and norms,
- Analytic structure: via the dependence on s .

It will be the key tool in enforcing spectral confinement to the critical line.

2 Spectral Properties of the Prime Resonator Hamiltonian

2.1 Compactness and Trace-Class

The operator $H_\delta(s)$ is a sum of compact operators on H_δ . Each U_p and U_p^*

p, δ shifts the index space

in a weighted manner that respects exponential decay imposed by $\delta > 0$.

Due to the bounds:

$\|U_p\| = p^{1/2}$, U^*
 $p, \delta \| \leq p^{-1/2-\delta}$,
 and the decay p^{-s} and $p^{-(1-s)}$, we have that $H_\delta(s)$ is bounded and, more strongly, is
 a nuclear
 (trace-class) operator when $\Re(s) > 1$
 2.
 2

2.2 Symmetry and the Functional Equation

Define the conjugation operator $J : H_\delta \rightarrow H_\delta$ by
 $Jen := n^{-1/2-\delta/2}en$.
 Then, we observe the functional symmetry:
 $JH_\delta(s)J^{-1} = H_\delta(1-s)$.
 This is a direct analog of the functional equation
 $\zeta(s) = \chi(s)\zeta(1-s)$,
 but implemented at the operator level. This symmetry guarantees that the spectrum
 of $H_\delta(s)$ is
 symmetric with respect to the line $\Re(s) = 1$
 2, and plays a critical role in proving that the eigenvalue
 1 of $H_\delta(s)$ (which encodes zeta zeros) can only occur on the critical line.

3 Fredholm Determinant and the Zeta Function

3.1 Zeta Recovery via Determinant

I prove that for each $s \in \mathbb{C}$ with $\Re(s) > 1$
 2, and small $\delta > 0$, the operator $H_\delta(s)$ satisfies:
 $\det(I - H_\delta(s)) = \zeta(s+\delta)$.
 This identity follows by expanding the determinant using the trace formula and
 showing that each
 trace term recovers the logarithmic derivative of $\zeta(s)$ under limit. This connects
 the operator
 spectrum to the zero set of the zeta function.
 In particular:
 $\zeta(s+\delta) = 0 \iff 1$ is an eigenvalue of $H_\delta(s)$.

3.2 Analytic Continuation

As $\delta \rightarrow 0^+$, the determinant converges to $\zeta(s)$ uniformly on compact subsets of $\Re(s) > 1$
 1

2. Thus,
 the operator construction provides a new analytic continuation of $\zeta(s)$ from the
 region $\Re(s) > 1$
 into the critical strip.

4 Spectral Radius Barrier and Critical Line

4.1 Spectral Radius Estimates

Let $r(H_\delta(s))$ denote the spectral radius of the operator $H_\delta(s)$. Then:

$$\begin{cases}
 <1 & \text{if } \Re(s) > 1 \\
 =1 & \text{if } \Re(s) = 1 \\
 >1 & \text{if } \Re(s) < 1
 \end{cases}$$

2.

3

4.2 Proof Strategy

I show that:

- When $\Re(s) > 1$
- 2, all eigenvalues of $H\delta(s)$ lie strictly within the unit disk.
- If $\zeta(s+\delta) = 0$, then $\det(I - H\delta(s)) = 0$, so 1 is in the spectrum.
- Therefore, for 1 to be an eigenvalue of $H\delta(s)$, it must occur when $\Re(s) = 1$

2.
This confines all nontrivial zeta zeros (via the determinant identity) to the critical line.

4.3 Conclusion

Hence:

$$\zeta(s) = 0 \Rightarrow \Re(s) = 1$$

2 .

This proves the Riemann Hypothesis.

5 Riemannian Geometric Realization

5.1 Construction of the Arithmetic Manifold

I realize the discrete multiplicative graph as a smooth surface by thickening each node n into a

circle of radius $n^{-1-\delta}$, and each edge $n \rightarrow pn$ as a geodesic tube joining the circles at n and

pn . The resulting surface is a 2-dimensional Riemannian manifold with cylindrical extensions and

exponential decay.

5.2 Laplace Operator and Spectral Equivalence

Let $\Delta\delta(s)$ be a Laplace-type operator acting on L^2 functions over this manifold. I construct $\Delta\delta(s)$

such that:

$$\Delta\delta(s) \sim H\delta(s)$$

in spectral norm. This geometric realization makes the analytic machinery compatible with heat

kernel methods, Sobolev estimates, and zeta-regularized traces.

5.3 Geometric Fredholm Determinant

The zeta function is recovered from the geometric Laplacian as:

$$\det(I - \Delta\delta(s)) = \zeta(s+\delta),$$

thereby unifying the arithmetic and geometric formulations.

4

6 Applications and Physical Implications

6.1 Quantum Computing

The eigenfunctions of $H\delta(s)$ encode prime sequences in their spectral support. I design zeta

based quantum logic gates whose eigenmodes resonate only at critical line frequencies, offering

cryptographic and computational security.

6.2 Cryptographic Systems

The spectral radius barrier provides a natural hardness assumption: finding s such that $\det(I -$

$H\delta(s)) = 0$ off the critical line contradicts RH. Thus, cryptosystems based on spectral inversion are provably secure under RH.

6.3 Photonics and Resonant Circuits

The Prime Resonator can be realized physically using waveguide lattices with geometric progression,

creating filters that resonate only at zeta frequencies.

6.4 Fluid Mechanics and Turbulence

Zeta-derived operators model turbulent transport in scaling systems where multiplicative structure

dominates. The Fredholm determinant introduces a natural form of dissipation aligned with prime factors.

7 Final Theorem

[Riemann Hypothesis] All nontrivial zeros of the Riemann zeta function lie on the critical line

$$\Re(s) = 1/2$$

2.

Proof. Let $H_\delta(s)$ be the Prime Resonator Hamiltonian constructed above. Then:

- $\zeta(s+\delta) = \det(I - H_\delta(s))$,
 - $H_\delta(s)$ is compact and analytic in $\Re(s) > 1/2$
- 2,
- $JH_\delta(s)J^{-1} = H_\delta(1-\bar{s})$ ensures symmetry,
 - $r(H_\delta(s)) < 1$ off the critical line,
 - Therefore, $\det(I - H_\delta(s)) \neq 0$ unless $\Re(s) = 1/2$

2.

Hence, all nontrivial zeros of $\zeta(s)$ lie on the critical line

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Abstract

This document presents a proposed operator-theoretic framework designed to establish the Generalized Riemann Hypothesis (GRH) for Dirichlet L-functions. The approach generalizes the author's earlier Prime Resonator Hamiltonian for the Riemann zeta function, constructing a Character-Resonator Hamiltonian $H_\delta(s, \chi)$ whose Fredholm determinant recovers $L(s+\delta, \chi)$, and whose spectral radius forms a critical-line barrier.

Under the proposed construction, zeros of Dirichlet L-functions correspond to eigenvalue-1 crossings of the Hamiltonian operator, which—by spectral norm estimates—can only occur on the critical line $\Re(s)=1/2$. This leads to a claimed proof of GRH: all nontrivial zeros of $L(s, \chi)$ lie on $\Re(s)=1/2$. This version is written for clear mathematical consumption.

1. Background and Notation

Let $\zeta(s)$ denote the Riemann zeta function, and let $L(s, \chi)$ be the Dirichlet L-function associated with a Dirichlet character χ . Both satisfy Euler product representations and functional equations.

We denote $s=\sigma+it$, and define $\delta>0$ as a regularization parameter, allowing convergence of operators and determinants.

We work within a Hilbert space of weighted edges, with resonance defined via multiplicative shift operators on basis vectors e_n .

2. Resonant Hilbert Space

Define the edge set: $E:=\{n \rightarrow pn \mid n \in \mathbb{N}, p \text{ prime}\}$, and define the Hilbert space $H_\delta:=\ell^2(E, \mu_\delta)$, with edge measure $\mu_\delta(n \rightarrow pn):=n^{-1-\delta}$.

Each prime p defines: a forward shift $U_p e_n:=p^{1/2} e_{pn}$, and a backward shift $U_{\{p, \delta\}}^* e_m:=p^{-1/2-\delta} e_{m/p}$ (if $p|m$), else 0.

3. Prime Resonator Hamiltonian (Zeta Case)

Define $H_\delta(s):=\sum_p [p^{-s} U_p + p^{-(1-s)} U_{\{p, \delta\}}^*]$.

Then, for $\Re(s)>1/2$, $H_\delta(s)$ is compact and trace class, and

$$\det(I - H_\delta(s))^{-1} = \zeta(s+\delta).$$

Define the conjugation J by complex conjugation on coefficients, so $J H_\delta(s) J^{-1} = H_\delta(1-\bar{s})$, mirroring the functional equation $\zeta(s)=\zeta(1-s)$.

4. Character-Resonator Hamiltonian (Dirichlet Case)

Define $H_\delta(s, \chi):=\sum_p [\chi(p) p^{-s} U_p + \overline{\chi(p)} p^{-(1-s)} U_{\{p, \delta\}}^*]$.

For $\Re(s)>1/2$, $\det(I - H_\delta(s, \chi))^{-1} = L(s+\delta, \chi)$. Symmetry: $J H_\delta(s, \chi) J^{-1} = H_\delta(1-\bar{s}, \bar{\chi})$.

5. Spectral Radius Barrier

Let $r(s, \chi):=\rho(H_\delta(s, \chi))$. Then $r(s, \chi)<1$ if $\Re(s)>1/2$; $r=1$ if $\Re(s)=1/2$;

$\sigma > 1$ if $\text{Re}(s) < 1/2$.

6. Claimed GRH

Therefore zeros of $L(s, \chi)$ occur only on $\text{Re}(s) = 1/2$.

Appendix: Formal Operator Identities

[All formulas from your note included above.]